

RESEARCH ARTICLE | NOVEMBER 20 2025

Study of the properties of nanoconfined EMIMBF₄ using polarizable molecular dynamics

Special Collection: [Chemical physics of ionic transport in solvents, polymers, and near interfaces](#)

Ángel Míguez-Roel ; Martín Otero-Lema ; Raúl Lois-Cuns ; Hadrián Montes-Campos  ; Trinidad Méndez-Morales ; Luis M. Varela 



J. Chem. Phys. 163, 194708 (2025)

<https://doi.org/10.1063/5.0293683>



Articles You May Be Interested In

Synthesis and characterization of ionic liquid (EMImBF₄)/Li⁺ - chitosan membranes for ion battery

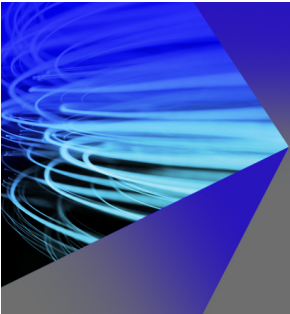
AIP Conf. Proc. (September 2015)

Ion conduction and relaxation mechanism in ionogels embedded with imidazolium based ionic liquids

J. Appl. Phys. (October 2019)

Mass loading vs electrochemical performance of battery supercapacitor hybrid

AIP Conf. Proc. (February 2025)



AIP Advances

Why Publish With Us?



21DAYS
average time
to 1st decision



OVER 4 MILLION
views in the last year



INCLUSIVE
scope

[Learn More](#)



Study of the properties of nanoconfined EMIMBF₄ using polarizable molecular dynamics

Cite as: J. Chem. Phys. 163, 194708 (2025); doi: 10.1063/5.0293683

Submitted: 29 July 2025 • Accepted: 27 October 2025 •

Published Online: 20 November 2025



Ángel Míguez-Roel, , Martín Otero-Lema, , Raúl Lois-Cuns, , Hadrián Montes-Campos, ^{a)} 
Trinidad Méndez-Morales, , and Luis M. Varela ^{b)} 

AFFILIATIONS

Grupo de Nanomateriais, Fotónica e Materia Branda, Departamento de Física de Partículas, Universidade de Santiago de Compostela, Campus Vida s/n, E-15782 Santiago de Compostela, Spain and Instituto de Materiais (iMATUS), Universidade de Santiago de Compostela, Avenida do Mestre Mateo 25, E-15782 Santiago de Compostela, Spain

Note: This paper is part of the Special Topic, Chemical physics of ionic transport in solvents, Polymers, and Near interfaces.

^{a)}Author to whom correspondence should be addressed: hadrian.montes@usc.es

^{b)}Electronic mail: luismiguel.varela@usc.es

ABSTRACT

We present a molecular dynamics study of the structure and dynamic behavior of the ionic liquid 1-ethyl-3-methylimidazolium tetrafluoroborate under nanoconfinement using a polarizable force field. Two distinct geometries were explored: cylindrical confinement within carbon nanotubes and planar confinement between graphene-like nanoslits, covering a range of confinement sizes. Our results reveal that the geometry and curvature of the confining region play a fundamental role in determining ion layering, molecular orientation, and diffusion. In planar nanoslits, highly ordered interfacial structures form regardless of pore width, while in carbon nanotubes, curvature suppresses such ordering and enhances ionic mobility near the interfaces. For a specific nanotube size, an anomalous behavior emerges where the combination of confinement geometry and pore dimensions promotes the formation of stable ionic layers, which drastically reduce diffusivity. These findings provide insights into the design of electrochemical devices and the optimization of ion transport in confined ionic media.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0293683>

I. INTRODUCTION

Ionic liquids (ILs) are salts composed entirely of ions that are in the liquid state below 100 °C. Typically formed by combining a bulky organic cation with either an organic or inorganic anion, ILs exhibit a suite of unique physicochemical properties that distinguish them from conventional molecular solvents and make them attractive as next-generation materials for a wide range of technological and environmental applications.^{1–4} Among their most notable properties are high thermal and chemical stability,⁵ negligible vapor pressure and non-flammability,⁶ wide electrochemical windows,⁷ and tunable solvation properties.⁸ The latter arises from the virtually limitless combinations of cations and anions, allowing for the design of task-specific ILs tailored to meet the requirements of specific applications. These properties have positioned ILs as green alternatives to volatile organic solvents in sustainable chemistry,⁹ reaction media in homogeneous and heterogeneous catalysis,^{10,11} separation and extraction processes, and chromatography.^{12,13}

Particularly prominent is the emergence of ILs in electrochemical technologies. Due to their high ionic conductivity and wide electrochemical and thermal stability windows, ILs have gained increasing attention as electrolytes in energy storage and conversion devices such as dye-sensitized solar cells,^{14,15} supercapacitors,¹⁶ and rechargeable batteries.¹⁷ Their potential to enhance safety, improve performance, and extend the lifespan of these systems has sparked extensive research into their integration into lithium-ion, sodium-ion, and other next-generation battery technologies.^{18,19}

In these electrochemical devices, a deep understanding of the behavior of ILs near interfaces is essential. The performance of ILs often depends critically on interfacial properties such as ion arrangement, layering effects, and surface-specific dynamics.^{20,21} These phenomena can significantly alter the physicochemical properties of ILs compared to those in the *bulk*.²² Furthermore, achieving high current densities in electrochemical devices requires the use of electrode materials that incorporate hierarchical porosity.^{23–25} Therefore, very frequently ILs are employed under conditions of

nanoscale confinement, either within porous carbon-based electrodes, nanopores in metal–organic frameworks (MOFs), or silica-based structures.^{26–28} The interaction of ILs with such complex structures leads to confinement effects that strongly influence ion transport, structuring, and electrochemical stability. Therefore, a thorough understanding of ILs inside porous or nanostructured materials and their structural and dynamical properties in proximity to solid interfaces remains a crucial challenge in the field.^{29–31}

To perform this kind of analysis, it is often necessary to rely on computational methods, particularly molecular dynamics (MD) simulations. Despite the large number of studies aimed at characterizing both the structural and dynamic properties of ILs in nanoconfined environments,^{32–36} to the best of our knowledge, not many simulations employing polarizable force fields have yet been conducted in this context. This absence is noteworthy, as polarizable models offer a more realistic representation of electrostatic interactions. As a result, the predictions obtained using such force fields are particularly novel, especially in terms of dynamic properties. While non-polarizable MD simulations can provide qualitatively accurate insights, they tend to significantly underestimate ionic mobility, often predicting much slower dynamics than those observed experimentally.^{37,38} Moreover, although the transport properties of ILs in the *bulk* phase are relatively well characterized and modeled,^{39–42} this is not the case under nanoconfinement. These discrepancies highlight the critical need for polarizable models to achieve a more quantitatively accurate description of IL behavior under nanoconfined conditions.

In the present work, we report polarizable MD simulations for one of the most extensively studied aprotic ILs: 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]). A detailed analysis is performed to characterize both the structural and dynamic properties of this IL under varying nanoconfinement conditions. In particular, we conduct a comparative study of the effect of two distinct nanoconfinement geometries: single-walled carbon nanotubes (CNTs), representing cylindrical confinement and graphene-like nanoslits, modeled as parallel implicit walls with the density of graphene. The characteristic pore sizes explored in this work range from 1.8 to 6.0 nm, corresponding to the diameter in the case of CNTs and the inter-wall spacing for nanoslits. This range allows us to investigate the transition from strong (“true”) nanoconfinement to moderate or mesoscopic confinement, where bulk-like behavior gradually begins to emerge. In addition, a 12 nm nanoslit is included to complete the dynamic analysis and to provide a clear reference for near-bulk conditions.

Section II outlines the methodological framework and simulation methods adopted in this study, while additional information regarding specific procedures can be found in the [supplementary material](#). In Sec. III, the results are then introduced and discussed. A summary of the most significant conclusions is provided in Sec. IV.

II. SIMULATION METHODS

This work employs polarizable molecular dynamics (MD) simulations to explore the behavior of the IL [EMIM][BF₄] under nanoconfinement at 370 K to enhance ion mobility and ensure sufficient statistical sampling for the analysis of dynamical properties. In addition, this temperature is well below the boiling point of [EMIM][BF₄] (estimated to be above 623 K^{43,44}), ensuring the

system remains in the liquid phase under ambient pressure. Two distinct confinement geometries were studied: cylindrical confinement within CNTs and planar confinement between implicit graphene-like walls, i.e., nanoslits. These two systems allow us to examine how geometry and dimensionality influence the structural and dynamic properties of confined ILs. For each geometry, a range of system sizes was considered to investigate size-dependent effects. In the case of CNTs, diameters corresponding to different chiral indices from 18 to 60 were selected: (18, 0), (20, 0), (22, 0), (25, 0), (30, 0), and (60, 0). For nanoslits, the distance between the two confining walls was set to 18, 20, 22, 25, 30, 60, and 120 Å, with a minimum of 300 ion pairs in each system to ensure sufficient sampling and statistical reliability. The structural parameters can be found in detail in the Tables I and II provided in the [supplementary material](#).

In the case of nanoslits, polarizable MD simulations were carried out using OpenMM version 7.6,⁴⁵ together with the CL&Pol polarizable force field developed by Goloviznina *et al.*,⁴⁶ which allows for an accurate description of both the system and its interactions. The workflow begins with the selection of a non-polarizable base force field, CL&P developed by Canongia Lopes *et al.*,^{47–52} followed by the assignment of atomic polarizabilities and the corresponding scaling factors for the Lennard-Jones terms, as described in Refs. 53 and 54. A detailed description of the construction of the polarizable force field is provided in the [supplementary material](#). Initial configurations were built using PACKMOL,⁵⁵ and the necessary input files for the polarizable simulations were generated with fftool⁵⁶ and pol_openmm.⁵⁷ Energy minimization was first performed with a tolerance of 10 kJ/mol. Afterward, stabilization runs were executed in the NpT ensemble for 10 ns. Subsequently, stabilization runs were performed in the NVT ensemble for 10 ns. The production runs were then carried out in the NVT ensemble for 15 ns.

To enforce the confinement, implicit carbon walls were used. Instead of introducing additional atoms, the potential was modified to account for the wall effects. Implicit walls are virtual barriers that are modeled indirectly through mathematical potentials, which effectively limit the movement of particles within a defined region, without the need for explicit representation of a physical boundary. In this study, a Lennard-Jones 10-4 wall potential was employed, which is mathematically expressed as

$$V_{\text{Walls}} = 8\pi\rho_w \left[\sum_j \epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{10z_j^{10}} - \frac{\sigma_{ij}^6}{4z_j^4} \right) + \sum_j \epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{10(z_j - z_w)^{10}} - \frac{\sigma_{ij}^6}{4(z_j - z_w)^4} \right) \right], \quad (1)$$

where interfaces are considered as a continuous 2D distribution of atoms with density $\rho_w = 38.18 \text{ nm}^{-2}$ (matching graphene density), and Lennard-Jones parameters $\sigma_i = 0.355 \text{ nm}$ and $\epsilon_i = 0.29288 \text{ KJ/mol}$, accounting for two walls (one at $z = 0$ and another at $z = z_w$). In this way, we choose to forego an explicit description of the interfaces. On the one hand, NpT equilibration cannot be performed with explicit walls. On the other hand, introducing them after an equilibration with implicit interfaces would slightly expand the box to match graphene periodicity and alter dynamics, so both NVT

equilibration and NVT production were carried out using implicit walls.

In the case of CNTs, the process is somewhat more complex and begins with the filling of the CNT with the IL. To achieve this, the method developed by Montes-Campos *et al.*⁵⁸ was followed. This step was performed using GROMACS 2019.5, employing the OPLS-AA force field. It is important to mention that, as the CNT is explicitly simulated, it is not possible to adjust the volume through an NpT equilibration of the IL inside the nanotube. Once the IL is inside the CNT, the simulation proceeds with OpenMM version 7.6 again. First, energy minimization is carried out with a tolerance of 10 kJ/mol, followed by a short 5 ns NVT equilibration using the non-polarizable CL&P force field. Following this brief equilibration, the final configuration is separated into the IL and the graphene wall. The IL force field is then polarized using `pol_openmm`. Once polarization is applied, the non-polarized CNT walls are incorporated, and a procedure practically identical to that employed for the nanoslits is followed: a new energy minimization is performed using the same convergence tolerance, followed by a 10 ns equilibration in the NVT ensemble and a subsequent 15 ns production run, also in the NVT ensemble.

Furthermore, with the aim of comparing the dynamical properties with those of the system without any confinement, separate *bulk* simulations were carried out. Four independent *bulk* simulation boxes were constructed with PACKMOL, each containing 400 ion pairs, and the CL&Pol force field was employed. The input files required for the polarizable simulations were generated using `fftool` in combination with `pol_openmm`. Energy minimization was subsequently carried out, employing a convergence tolerance of 10 kJ/mol. A NpT equilibration at 370 K and 1 bar was first conducted for 5 ns, followed by an NVT equilibration at 370 K for 10 ns. Finally, the diffusion coefficients were obtained from a 15 ns NVT production run at 370 K.

For CNTs, nanoslits, and *bulk* systems, a time step of 1 fs was maintained throughout all simulations, while hydrogen bonds were kept fixed using the SHAKE algorithm.⁵⁹ Throughout all MD simulations, temperature was controlled using a temperature-grouped Nosé-Hoover thermostat.⁶⁰ The translational motion of the center of mass was thermalized at 370 K with a collision frequency of 10 ps⁻¹, while the temperature of Drude particles (if present in the simulations) was maintained at 1 K with a collision frequency of 40 ps⁻¹. For the NpT ensemble simulations of slit systems, pressure was regulated at 1 bar using an anisotropic Monte Carlo barostat,^{61,62} with the simulation box dimension in the Z-direction kept fixed to match each slit size. Although the barostat is technically anisotropic, it operates isotropically in the XY plane, meaning that pressure is coupled equally in X and Y but not in Z. Meanwhile, for the *bulk* NpT equilibrations, an isotropic Monte Carlo barostat was employed. To ensure the stability of the trajectories, a hard wall constraint was imposed on the maximum distance between the core and its Drude particle, limiting it to 0.2 Å. Electrostatic interactions were handled using the smooth particle mesh Ewald method with a real-space cutoff of 12 Å and an error tolerance of 10⁻⁵. In addition, van der Waals interactions were truncated above the same cutoff. Moreover, to prevent potential artifacts arising from periodic images, a vacuum gap was introduced in the simulations for nanoslits and CNTs. In particular, for nanoslits smaller than 3 nm, the vacuum region was set to twice the simulation box length, whereas for longer nanoslits, a

spacing equal to the box length was employed. For the CNT systems, the nanotubes were positioned at a corner of a 10 × 10 nm² simulation box in the (x, y) plane, while the box length along the z axis corresponded to the nanotube length, leaving a substantial vacuum region within the box to achieve the same objective. In addition, the diffusion coefficients of the ionic species were calculated using the KUTE package,⁶³ which estimates them through the Green-Kubo formalism.

Finally, all systems were simulated at constant wall charge, with a fixed uniform surface charge set to zero, without permitting any point-to-point fluctuations. On the one hand, constant-charge simulations yield poorer structural and dynamical results compared to constant-potential methods, especially under strong nanoconfinement,⁶⁴ as they do not account for electrode polarization. In general, four main approaches exist to implement metallic electrode polarization: image charge methods, solving the Poisson equation on a surface mesh to obtain the induced surface charge density, using mobile charges, or employing fluctuating charges.⁶⁵ However, most of these methods are straightforward to implement only for planar electrodes.⁶⁶ In OpenMM, the most straightforward way to implement electrode polarization is via the image charge method.^{67,68} While this method is easily implemented for planar walls (as in slits), it is not the case for CNTs, where the cylindrical geometry prevents a trivial image charge treatment. This is because the image of a point charge in a cylinder is not another point charge, as in the planar case, but rather a continuous charge distribution that is not easily determined analytically and goes beyond the scope of this study. Nevertheless, the use of the constant charge method for nanoslit systems remains advantageous: it is computationally less demanding and allows for consistent comparison with CNT systems. In addition, for slits, preliminary analyses using the image charge method for the most strongly confined system showed no significant differences between the results obtained with both approaches. The results obtained for the 18 Å wide slit using the constant potential method compared to those employing the constant charge one are included in the [supplementary material](#). Therefore, for consistency and computational efficiency, all simulations were conducted at constant charge.

III. RESULTS

A. Density profiles

The main structural changes resulting from variations in the size of the nanoconfinement were analyzed. To investigate these effects, the number density of each species was represented along the characteristic size of each geometry. For CNTs, this size corresponds to the radial coordinate, while for nanoslits it corresponds to the axis perpendicular to the interface (z axis). The density profiles were normalized such that their total integral equals unity. The results are presented in [Fig. 1](#). Density profiles were obtained from the positions of all atoms within each molecule, excluding the associated Drude particles.

Initially, a common trend is observed: in the most confined systems, ionic density oscillations arise, corresponding to charge oscillations. As the degree of confinement decreases, new ionic layers form farther from the interfaces. In the case of CNTs, these layers adopt concentric shell structures, while for the nanoslits planar layers are formed. This behavior continues up to confinements of

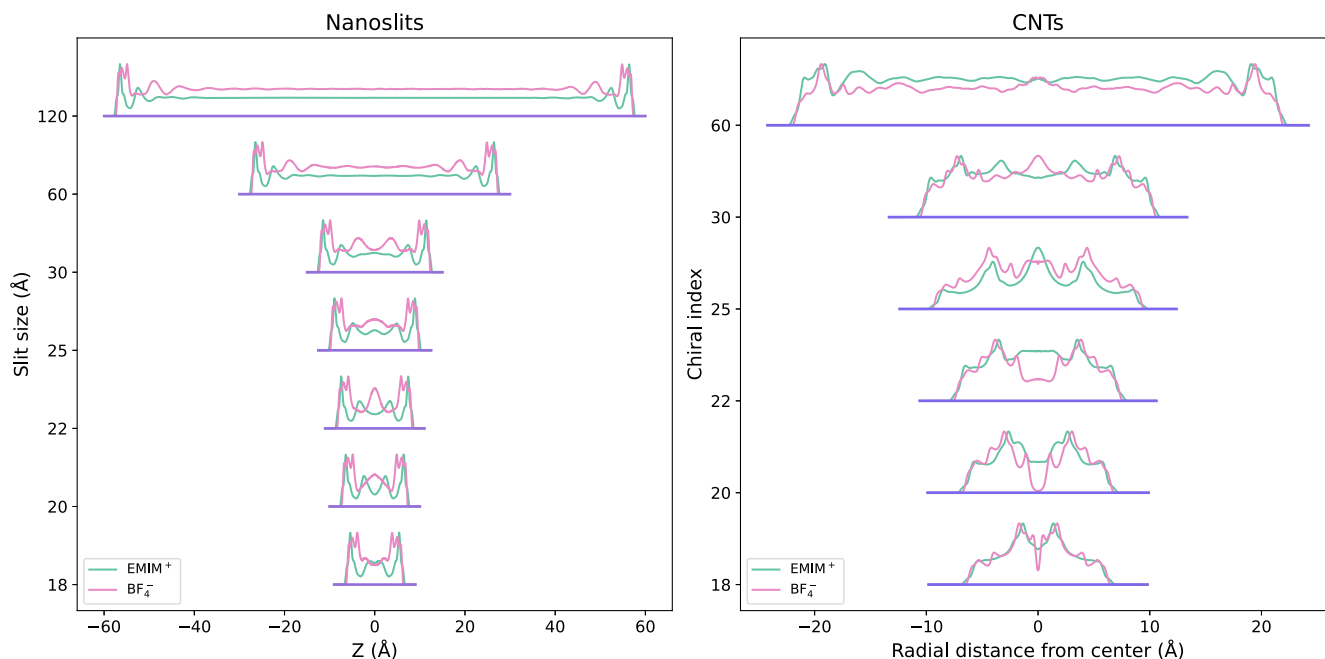


FIG. 1. Density profiles of each species along the symmetry axis, normalized to unit area. The y axis indicates the slit size in Å and the chiral index of the CNT for the corresponding case.

~30 Å, and for wider systems, a central *bulk*-like region emerges. It should be noted that, in the case of CNTs, this central region does not behave exactly as expected for a simulated *bulk*. This discrepancy is attributed to the fact that the density in this region does not yet correspond to that of a simulated *bulk* phase.

Another important observation is that the ionic distribution forming the electrical double layer near the interface remains largely unaffected by the slit size. In contrast, this behavior differs significantly in CNTs. This discrepancy stems from the fact that planar surfaces provide a uniform and consistent interaction across their entire area. As a result, the laminar nature of slit confinement ensures both the density profile and the double layer structure remain essentially constant despite variations in pore thickness.

Conversely, in CNTs, the curvature of the confinement plays a crucial role, especially at smaller diameters, profoundly influencing ion organization. In narrow CNTs, the cylindrical geometry severely restricts the number of concentric layers that can form and intensifies interactions between ions and the inner wall. This represents a more stringent confinement, compelling the IL to interact with a greater number of carbon atoms on the wall. Such steric spatial constraints weaken as the nanotube radius increases, allowing electrostatic interactions between cations and anions to become dominant. This shift likely accounts for the distinct structural arrangements observed in larger-diameter nanotubes.

As the CNT diameter grows, the curvature of the wall decreases accordingly, which reduces IL-wall interactions in the layers adjacent to the surface. Consequently, the electrical double layer evolves with the CNT radius from a strongly confined state that restricts the IL approach to the interface to a regime where the IL exhibits

enhanced affinity for the walls, resulting in the formation of thinner, denser ionic layers closer to the surface.

B. Orientational order

These results are clearly supported by observations of the orientational order of the species in the region near the interface. In particular, the probability density distribution of the cosine of the angle between a vector normal to the interface (pointing toward the *bulk*) and a characteristic vector for each molecule is computed. For EMIM⁺, this vector originates at the geometric center of the imidazolium ring and points perpendicular to the ring plane. For BF₄⁻, the vector aligns along one of the B-F bonds. For better understanding, please see the representation of the molecules and the characteristic vectors in Fig. 4 of the [supplementary material](#). These distributions are calculated by evaluating the probability of molecular alignment in a given direction at a fixed z-coordinate for nanoslits, and at a fixed radial distance for CNTs, integrating over the 4–7 Å region closest to the interface, which approximately corresponds to the first ionic layer near the interface, with the exact range varying slightly depending on the specific system. To define these integration regions, a heat map of orientation probabilities as a function of distance from the interface is generated, as shown in Fig. 2 for EMIM⁺ in some specific systems. In the case of nanoslits, this interfacial region typically consists of one cationic and one anionic layer, both exhibiting a well-defined orientational order. In contrast, in CNTs, it adopts a concentric annular structure comprising two distinct layers of IL that still retain orientational organization close to the interface. This notable difference in the stratification of the ionic layers is clearly illustrated in Fig. 2 for EMIM⁺. As previously

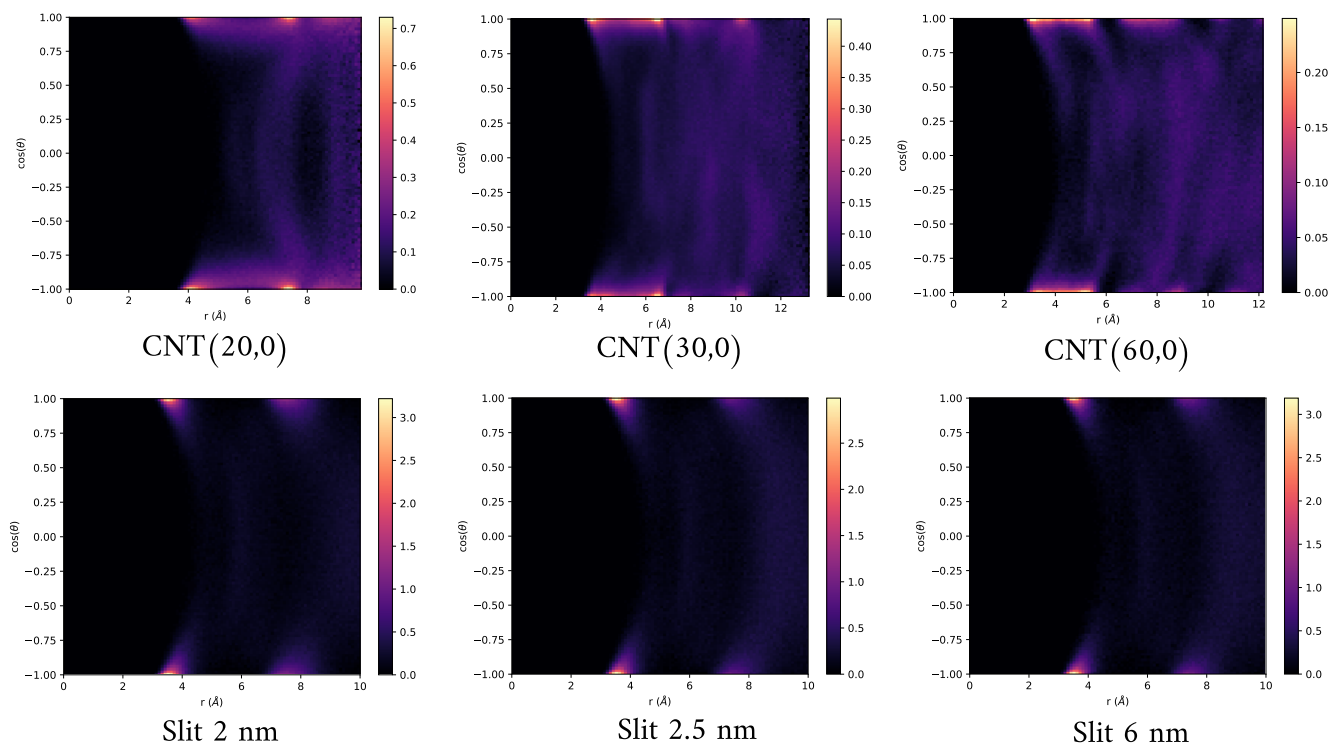


FIG. 2. Heat map of the orientations of EMIM⁺ near the graphene wall. The top row displays results for CNTs with chiral indices 20, 30, and 60 from left to right. The bottom row shows the corresponding results for nanoslits with widths of 20, 25, and 60 Å also from left to right. The slit sizes are selected to better match the CNT diameters associated with the chosen chiral indexes, thereby improving the comparison. Note that the graphene wall is placed at 0 Å. The color scale corresponds to the square root of the normalized orientation probability density, employed to enhance the visibility of the orientational structure.

discussed and evident from this representation, the interfacial IL layers in nanoslits maintain a consistent stratification regardless of the slit width. In contrast, for CNTs, the layers become narrower, more sharply defined, and appear progressively closer to the nanotube wall as the diameter increases.

The results of the aforementioned integration are shown in Fig. 3. As anticipated, the orientational structure in slits remains independent of the slit width, consistent with the previously observed density profiles. In these systems, the EMIM⁺ cation aligns its imidazolium ring almost entirely parallel to the interface, whereas BF₄[−], being a smaller and more symmetric anion, exhibits a broader distribution of intermediate orientations. Notably, the four peaks observed in the orientational distribution of BF₄[−] correspond to only two distinct orientations due to the molecular symmetry and the specific definition of the vector used: alignment of either a B–F bond or a face of the BF₄[−] tetrahedron toward the interface.

In contrast, for CNTs, the orientational behavior is notably different. In the case of EMIM⁺, the molecule tends to align its planar ring parallel to the inner wall for small-diameter nanotubes, but the high curvature of the nanotube prevents full alignment. As the CNT diameter increases, curvature decreases and the cations orient completely parallel to the surface. A similar trend is observed for BF₄[−], although less pronounced due to its smaller size. In addition, in CNTs, BF₄[−] tends to lose the preferential orientation seen in planar systems; it only shows a tendency to place a BF₄[−] tetrahedron

face toward the interface and a single B–F bond pointing toward the center of the nanotube.

C. Dynamic properties

Next, the dynamic properties of the systems are investigated. To this end, the diffusion coefficients of each species were computed. To account for potential spurious hydrodynamic effects on diffusion parallel to the confining walls, we applied the corrections proposed in Ref. 69 for the parallel diffusion component in nanoslits. Regarding cylindrical geometries, to the best of our knowledge, no comparable studies addressing the characterization of these artifacts have been reported in the literature. While a systematic investigation of such effects lies beyond the scope of the present work, the geometry and the pronounced degree of confinement in nanotubes suggest that these spurious hydrodynamic contributions are unlikely to play a significant role.

Diffusion under nanoconfinement within CNTs is first examined. The spatial components of the diffusion coefficient are separated to enable a more detailed analysis. The behavior is illustrated in Fig. 4. It is worth noting that diffusion in the nanotube is significantly slower than in the slits and the *bulk* (as it will be shown). Furthermore, diffusion in the *xy*-plane is generally much greater than in the *z*-direction for both ions. This indicates that the confinement restricts axial transport more severely than transverse motion. However, both the transverse and axial diffusion

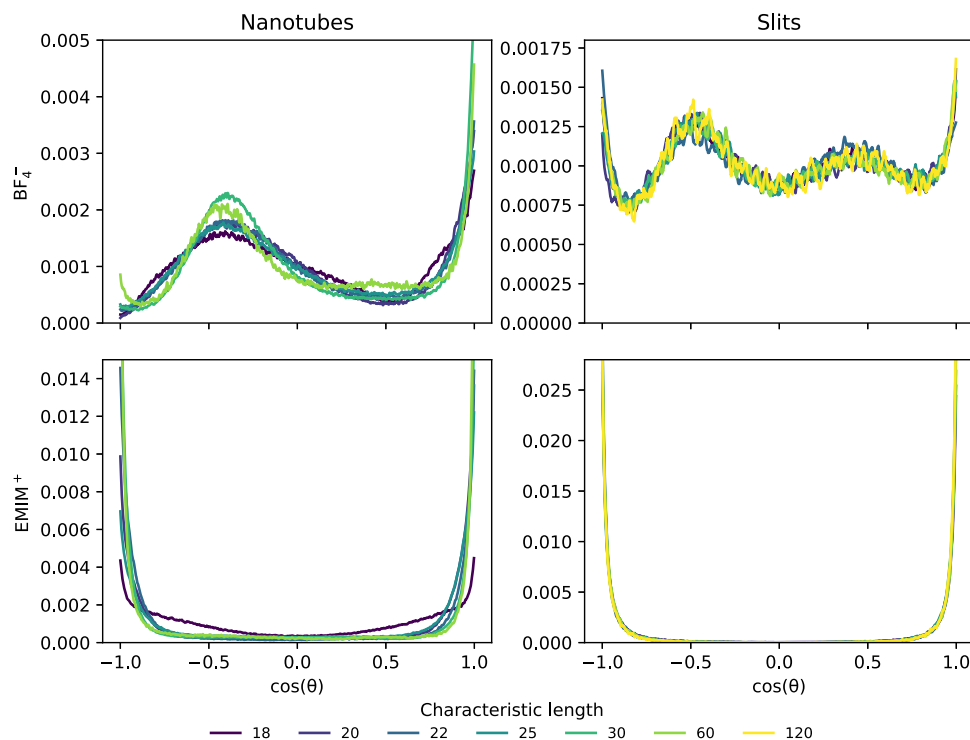


FIG. 3. Probability distributions of the cosine of the angle between the vector normal to the interface and the characteristic molecular vector (BF_4^- top, EMIM^+ bottom). The left panels correspond to CNTs and the right panels to nanoslits. The characteristic length is expressed in angstroms for slits (slit width) and as the chiral index for CNTs.

components exhibit a similar trend, differing only in their absolute values. Therefore, a general trend is observed in the total diffusion behavior: diffusion increases at small nanotube diameters and then decreases as the diameter grows, eventually reaching an almost constant value for chiral indices above ~ 30 . In the case of the CNT(60, 0), a *bulk*-like region emerges in the central area of the nanotube. However, the diffusion observed is approximately two orders of magnitude slower than that of the simulated *bulk* despite the majority of molecules being located in this central region. This effect arises from the local density in the central region being $\sim 22\%$ higher than that of the corresponding simulated *bulk* system. Furthermore, the CNT(30, 0) exhibits anomalous behavior in its diffusion, with a clearly reduced diffusivity.

The phenomenon of improved diffusion promoted by reduced nanotube diameter has been previously reported in similar systems. For instance, Ghoufi *et al.*,⁷⁰ using MD simulations, showed that the diffusion coefficient of the cation $[\text{C}_4\text{MIM}]^+$ confined within a CNT of radius ≈ 13 Å is higher than that in larger-radius nanotubes. In their case, this behavior was attributed to reduced friction: as the nanotube radius decreases, surface friction is lowered, allowing ions to move more freely. These MD results are consistent with some experimental findings. Berrod *et al.*,⁷¹ using NMR techniques, measured enhanced mobility of imidazolium ions ($[\text{OMIM}]^+$) under CNT confinement. Similar observations have also been reported in other systems involving neutral molecules such as water.⁷² In that case, the increase in diffusion was associated with the disruption of the hydrogen-bond network due to enhanced nanoconfinement and stronger water-wall interactions. Despite the use of an aprotic IL that lacks a hydrogen-bond network, confinement, and especially its geometry, can still have a substantial impact on the stability of the

charge layers forming the electrical double layer in the vicinity of the IL-wall interface.

A key factor influencing the behavior and trends observed could also be the IL density. In fact, plotting the total diffusion coefficient against the mass density of the IL in each CNT reveals an almost linear relationship, as seen in Fig. 5. It is worth noting that the density values were calculated using the cylindrical volume formula $V = \pi r^2 h$, where h is the length of the CNT and r is the distance from the center of the nanotube to the center of the carbon atoms forming the wall, reduced by 2.0 Å. This subtraction accounts for the region that the IL cannot occupy due to van der Waals repulsion of graphene, defining an effective or “accessible” volume that can realistically be filled by the IL.³¹ However, as previously discussed, the actual spatial distribution of the IL near the interface varies with the chiral index of the CNT. In general, especially for narrow nanotubes, the IL occupies a volume smaller than this defined accessible region.

Figure 5 provides several insights. First, a clear inverse correlation is observed between the total diffusion coefficient and the IL density inside the nanotube. This observation raises the question of whether the enhanced diffusion observed in narrower CNTs is primarily a result of confinement geometry or simply a consequence of reduced density, which naturally facilitates greater molecular mobility. Second, the trend aligns with known behaviors in $[\text{EMIM}][\text{BF}_4]$ mobility under varying density conditions.^{42,73} Finally, the data confirm a notable deviation in the case of the CNT(30, 0), which exhibits an almost negligible diffusion coefficient, substantially lower than expected and clearly departing from the general trend, as previously observed in Fig. 4. Several of the issues identified at this point will be explored in greater depth later on. Nevertheless, before proceeding

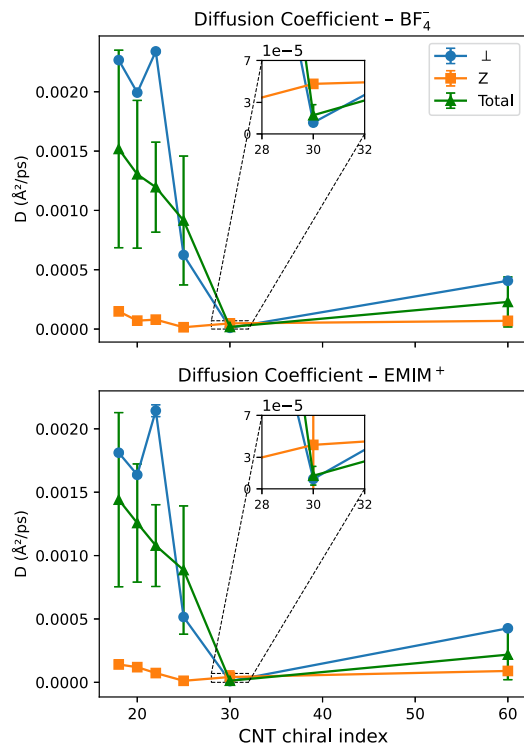


FIG. 4. Diffusion coefficients for each species confined inside CNTs as a function of their chiral index. The blue curve corresponds to diffusion in the xy-plane; the orange curve represents axial (z-direction) diffusion; and the green curve indicates the total diffusion coefficient. The top panel shows results for BF_4^- , while the bottom panel corresponds to EMIM^+ . A zoom-in on the results for the CNT(30, 0) is included to highlight the behavior in that specific case; the axis units are the same as in the main plot.

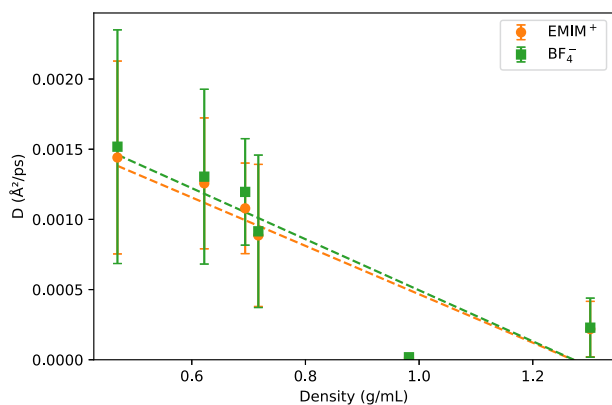


FIG. 5. Total diffusion coefficients for each species as a function of the total IL mass density in each CNT. The data points correspond to the calculated values, while the dashed line represents the linear fit.

with their analysis, Fig. 6 presents the diffusion coefficients obtained for the ionic species confined within the nanoslits.

In this case, notably different results are obtained. On the one hand, although diffusion in the xy-plane is greater than in the axial

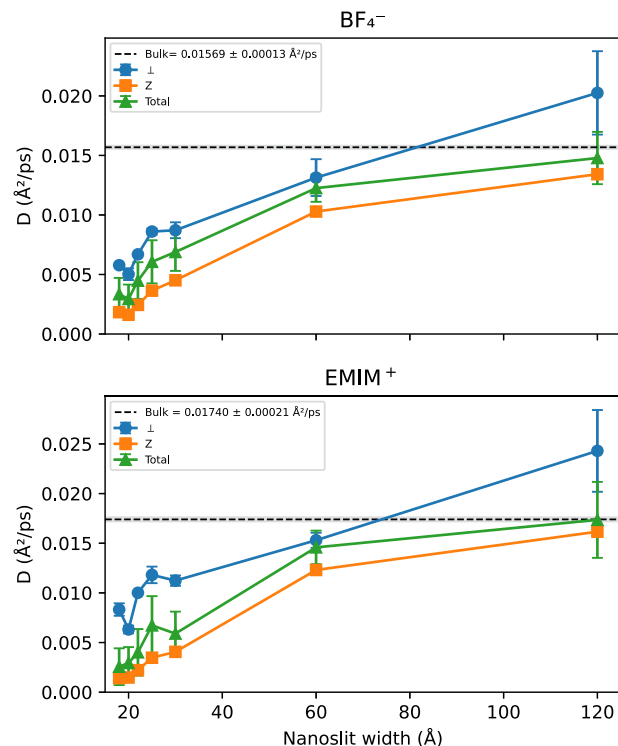


FIG. 6. Diffusion coefficients for each species confined inside nanoslits as a function of the nanoslit width. The blue curve corresponds to diffusion in the xy-plane; the orange curve represents axial (z-direction) diffusion; and the green curve indicates the total diffusion coefficient. The top panel shows results for BF_4^- , while the bottom panel corresponds to EMIM^+ . Moreover, the *bulk* diffusion coefficient for each ion obtained from 4 replicas is presented in gray.

direction, the difference is not as pronounced as in CNTs as well as in CNTs, both the transverse and axial coefficients follow the same trend. Another relevant aspect is the global observed trend: unlike in CNTs, increasing the slit size along the z axis does not lead to a decrease in diffusion, but rather to an increase until values close to those of the *bulk* are reached. It is worth noting that for the 120 $\text{\AA}$ slit, we obtain slightly higher values for the transverse diffusion coefficient compared to the *bulk*. However, this value is associated with a high degree of uncertainty in its estimation. In contrast, more accurate values are obtained for both the diffusion coefficient along the Z-direction and the total diffusion coefficient, for which the *bulk* values are effectively recovered. The recovery of a large *bulk*-like central region for wider slits, where the density closely matches that of the *bulk*, means that while dynamics near the interfaces are affected by wall interactions, most molecules diffuse as if the slit walls were absent.

The reduced ionic mobility under stronger nanoconfinement is attributed to the planar geometry of these pores, which allows ions to arrange into highly stable charge layers near the interfaces. Similar results have been previously reported in related systems employing MD simulations.³¹ Furthermore, the contrast between the diffusion trends observed in CNTs and nanoslits has also been reported by other authors for different ILs.⁷⁰ This enables a clear distinction between the effects of the size of the confining region and those due

to its curvature. The results indicate that curvature of the confining region plays a crucial role preventing the formation of stable structures near the interface, unlike planar geometries, thereby increasing ionic mobility.

Once both trends have been compared and the influence of the confining geometry on the dynamic properties has been established, a more in-depth analysis of the questions that arise in the case of CNTs is now carried out. The first analyzed aspect is the notable loss of mobility inside the CNT(30, 0). As shown in the zoomed-in view of Fig. 4, the drastic reduction in the diffusion coefficient is primarily associated with a sharp decrease in transverse mobility. In contrast, the axial diffusion remains within the expected range, following the general trend. However, in the *xy*-plane, the diffusion becomes nearly negligible. To elucidate these observations, the structural properties were analyzed in order to uncover the origin of the suppressed transverse diffusion. Density profiles in Fig. 1 reveal that, for the CNT(30, 0), the density of the internal layers is maximized, and a third layer appears for the first time. This behavior is schematically illustrated in Fig. 7, where it is observed that while only two layers are present in narrower CNTs, a third layer emerges specifically in the case of the CNT(30, 0). From this representation, it can also be observed that for the CNT(20, 0), a transition occurs from a single layer to a two-layer structure of the IL. A similar effect is predicted for CNT(20, 0): a reduction in transverse diffusion, which increases again for the following CNT.

Such phenomena have been previously reported, for instance, in ethylammonium nitrate (EAN) confined in nanoslits,³¹ where the formation of a new layer upon increasing the slit width leads to local maximum in density. This occurs when the spacing between confining interfaces allows for densely packed ionic configurations,

thereby minimizing the available free volume. As a consequence, in the CNT(20, 0) and the CNT(30, 0), a particularly ordered, stable, and locally dense structure is formed, which restricts both interlayer transport and in-plane motion of the IL.

Nevertheless, this situation can be further clarified by examining the orientational order of the ions. To this end, the probability density distribution of the cosine of the angle between a vector normal to the interface (pointing toward the *bulk*) and the characteristic vector of each molecule is evaluated at a fixed radial distance. The entropy associated with this orientational structure is then computed as

$$S(r)/k_B = - \int p(r, \cos \theta) \ln(p(r, \cos \theta)) d \cos \theta - \ln(N_{\text{bins}}), \quad (2)$$

where $p(r, \cos \theta)$ is the probability of a given orientation at a given r and N_{bins} is the number of bins used to divide the space of possible orientations. The final term is included to ensure a zero value corresponds to the absence of orientational structuring. The presence of low-entropy wells in the resulting curve indicates the existence of ordering at specific radial places, whereas flat zero regions indicate a lack of orientational organization. Due to the definition of the angle, particles near the center of the nanotube do not have a well-defined orientation, so the presence of peaks in this region is generally considered a non-physical artifact.

The results presented in Fig. 8 reveal the emergence of long-range orientational order away from the wall for the CNT(25, 0) at ~ 8 Å from the first ionic layer near the wall. This phenomenon indicates a structural transition in the nanoconfined IL compared to CNTs of smaller diameter, specifically marked by the formation of a new IL concentric ring. This deep orientational ordering becomes more pronounced in the CNT(30, 0), where the appearance of a third IL layer leads to the development of a sharper low-entropy well. This effect is also evident in Fig. 2, where the orientational structure for

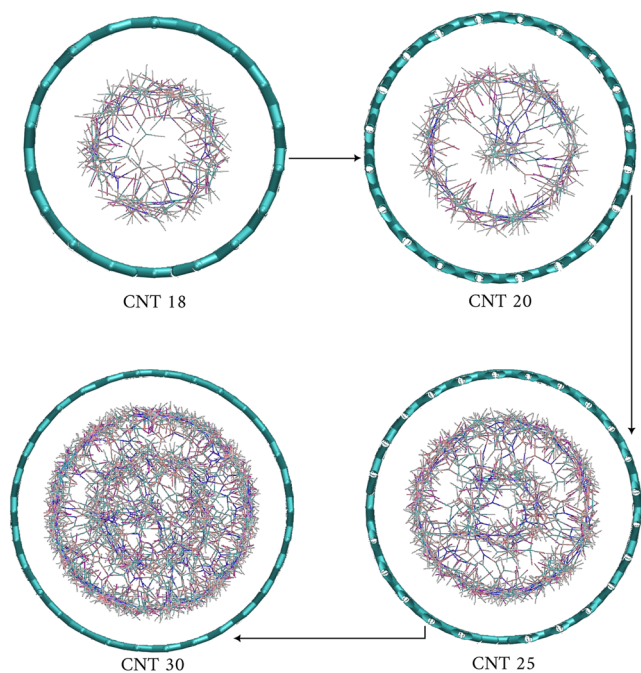


FIG. 7. Cross-sectional view of the concentric layers formed by the IL inside the CNT for increasing chiral indices in a clockwise direction.

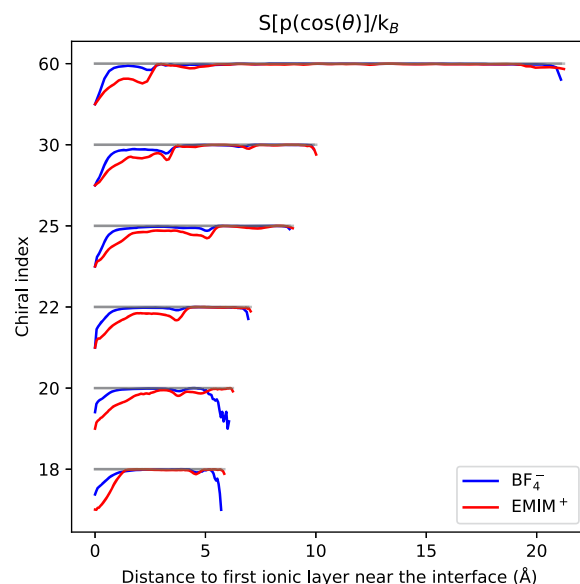


FIG. 8. $S[p(\cos(\theta))]/k_B$ as a function of the distance to the first IL layer formed near the interface for all the CNTs.

the CNT(30, 0) persists at ~ 10.5 Å (note that in Fig. 8 the distance is to first ionic layers, not the CNT wall) from the graphene wall, which is exactly the same distance at which the low-entropy well appears. This finding is particularly significant, as it provides further evidence for the existence of ordered and stable structures in CNT(30, 0), which may be responsible for the observed decrease in diffusion. Moreover, when considering only the three largest CNTs studied, the orientational order appears consistent across them. However, as the nanotube diameter increases and the degree of confinement decreases, the interfacial layers become progressively more compact. This relaxation of confinement also leads to a faster decay of the orientational order, preventing it from persisting deeper into the bulk.

In contrast, Fig. 9 presents the entropy profiles calculated for the nanoslits. Once again, the distinction between the two types of confinement becomes evident. In the case of nanoslits, additional IL layers emerge as the slit width increases, without significant changes in either the behavior or the position of the ordered layers (low-entropy wells) closest to the walls, those already present at smaller widths. Finally, for the 60 Å slit, a bulk-like region is recovered, and no significant orientational ordering is observed 10 Å away from the first ionic layer. Thus, it is worth noting that orientational ordering is now maintained in deeper layers from the interface, indicating a more stable and structured layering than that observed in the CNT. This provides further evidence of the marked differences in ionic structuring that arise depending on the curvature of the confining environment, as previously discussed.

On the one hand, all these results highlight the importance of the confining structure. On the other hand, the non-monotonic tendency of structural and dynamical properties observed in carbon nanotubes, in contrast to the monotonic behavior seen in slits, indicates that these properties are directly influenced by the emergence of new structures. Their formation is associated with changes

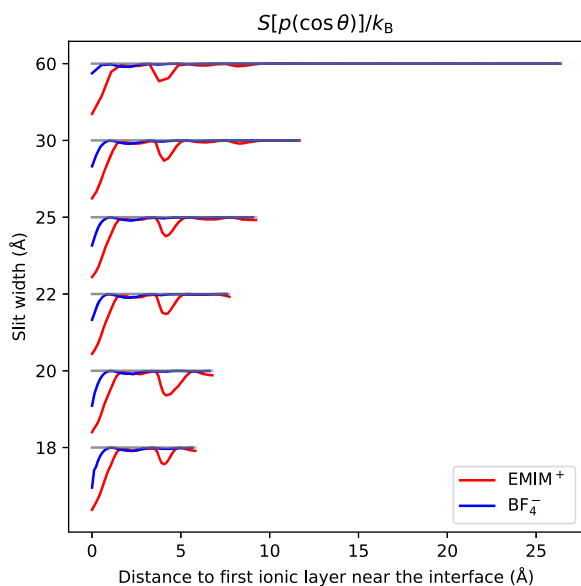


FIG. 9. $S[p(\cos \theta)]/k_B$ as a function of the distance to the first IL layer formed near the interface different slit sizes.

in the degree of confinement; for instance, increasing the CNT diameter allows the IL to form additional concentric layers. Thus, while monotonic tendencies can be identified for similar structural arrangements, the emergence of new configurations disrupts these trends.

Finally, the total diffusion will be analyzed as a function of position in the confined dimension, both for CNTs and nanoslits. This analysis not only enables a comparative study of how the different geometries of confinement affect ionic mobility in the interfacial layers but also provides insight into the relative importance of curvature vs IL density in governing ion diffusion within CNTs, addressing one of the questions raised earlier. To this end, the largest CNT and the 6 nm slit have been selected for analysis. Based on the density profiles obtained from these systems, a set of layers has been defined, to which molecules can be assigned. Molecules that remain within a given layer throughout the entire simulation are used to compute the diffusion coefficient in that specific region. Given the high diffusivity observed in the slit, the space has been divided into only three regions to ensure enough statistics: the two lateral regions, comprising the 15 Å closest to each interface, and the central 30 Å region, which behaves as a bulk-like domain. In contrast, due to the slower diffusion in CNTs, a finer division into more layers is possible, allowing for a more detailed analysis. The results are shown in Figs. 10 and 11.

Once again, markedly different trends emerge between the two confinement geometries. In the case of the slit, molecular mobility in the near-interface region is lower than in the bulk, a trend that has

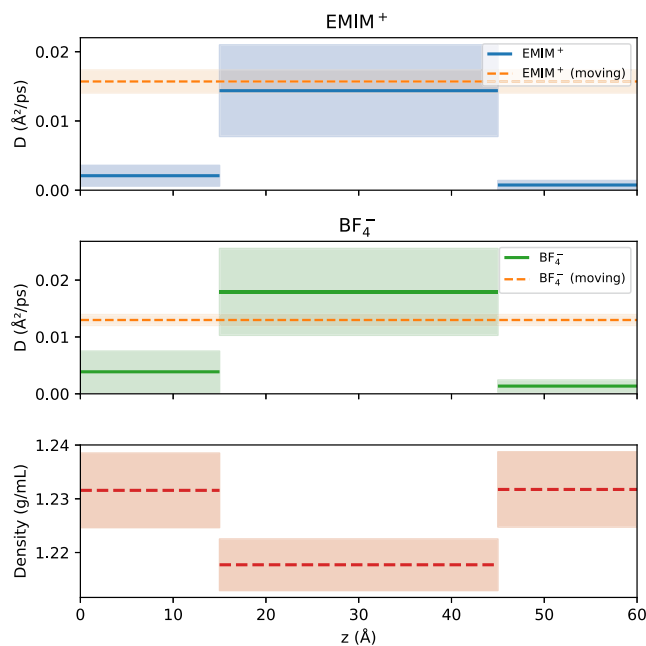


FIG. 10. Total diffusion coefficient for each layer as a function of the z -coordinate for the 6 nm slit. In the top panel, results for EMIM⁺ are shown: in blue for the defined layers and in dashed orange for the molecules that cannot be assigned to any specific layer. In the middle panel, results for BF₄[−] are shown: in green for the layers and in dashed orange for the molecules that cannot be assigned to a specific layer. In the bottom panel, the total IL mass density is plotted as a function of the axial distance.

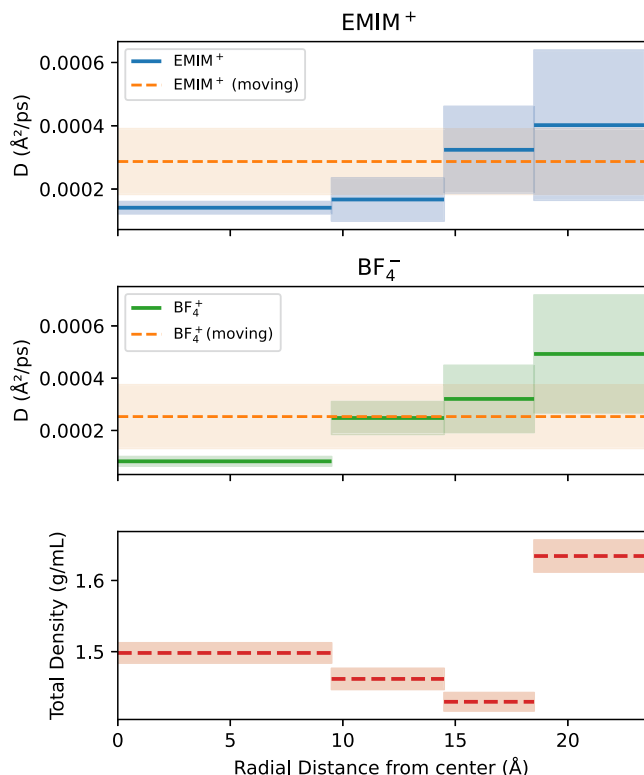


FIG. 11. Total diffusion coefficient for each layer as a function of the radial position for the CNT(60, 0). In the top panel, results for EMIM^+ are shown: in blue for the defined layers and in dashed orange for the molecules that cannot be assigned to any specific layer. In the middle panel, results for BF_4^- are shown: in green for the layers and in dashed orange for the molecules that cannot be assigned to a specific layer. In the bottom panel, the total IL mass density is plotted as a function of the radial distance.

been observed in previous studies of similar systems.³¹ This result is consistent with the evolution of the diffusion coefficients as a function of slit size, where stronger interactions with the walls favor the formation of stable layered structures, which limits the mobility of the ions. In addition, for the 60 Å slit, these interfacial layers exhibit higher ionic density, further contributing to this effect. On the other hand, for the CNT, diffusion in the central region, where ions are no longer arranged in concentric rings but instead occupy a *bulk*-like domain, is significantly lower. Interestingly, diffusivity increases as one moves toward layers closer to the nanotube wall. Moreover, it is evident that this trend cannot be directly attributed to the density of the layers, as the highest mobility is found in the densest one. Some MD studies have addressed this heterogeneity. Dick *et al.*⁷⁴ conducted charge-scaled MD simulations of pure $[\text{EMIM}][\text{BF}_4]$ and its mixtures with NaBF_4 , confined within a CNT. They found that atoms near the inner surface of the nanotube exhibited higher average velocities than those located at the center of the pore. In agreement with the trends observed when varying the characteristic size of the confining geometry, it is found that for nanoslits, highly stable and ordered charge structures form near the interfaces, which significantly limit ionic mobility. In contrast, for carbon nanotubes, the curvature prevents the formation of such ordered

layers, resulting in an increase in ionic mobility near the interface compared to the *bulk*-like region, despite this interfacial layer being denser.

IV. CONCLUSIONS

This study presents a comprehensive molecular dynamics investigation into the structural and dynamic behavior of the aprotic IL $[\text{EMIM}][\text{BF}_4]$ under nanoconfinement, employing polarizable force fields to achieve a realistic depiction of electrostatic interactions. The analysis was conducted across two distinct confinement geometries, cylindrical pores formed by single-walled carbon nanotubes and planar pores modeled as graphene-like nanoslits, covering a broad range of pore sizes, from strong nanoconfinement (about 1.8 nm the smallest systems) to near-*bulk* conditions (up to 12 nm in nanoslits).

The simulations revealed that confinement geometry plays a decisive role in determining both the structural organization and transport properties of the confined IL. In planar nanoslits, the IL exhibits highly stable and well-ordered interfacial structures, which remain robust regardless of slit width. These interfacial layers are characterized by pronounced ionic layering and a strong preferential orientation of ions near the interface. Conversely, in CNTs, the curvature significantly alters interfacial organization. Narrow CNTs hinder the formation of ordered layers and restrict the parallel alignment of the imidazolium ring due to geometric constraints. As the nanotube diameter increases, curvature effects diminish, leading to improved alignment and the gradual emergence of *bulk*-like behavior in the central region.

The emergence of new structural arrangements within confined environments has been found to significantly alter transport trends, breaking the expected monotonic behavior with pore size. These structural transitions, driven by changes in the degree of confinement, can drastically affect ionic mobility by inducing enhanced ordering and layering. A striking example of this behavior is observed in the CNT(30, 0), where a pronounced drop in transverse diffusion is detected. This anomaly is linked to the formation of a third, well-organized ionic layer inside the pore, which marks a structural transition that severely limits in-plane motion. Such results highlight how specific geometric configurations can give rise to unexpected transport properties due to the interplay between confinement and structural organization.

Diffusion dynamics further highlight the importance of the confinement. In CNTs, diffusion coefficients are enhanced near the interfaces and in narrower tubes, where structural disorder dominates and promotes greater ionic mobility. This behavior contrasts with that observed in planar slits, where diffusion is markedly reduced near the interfaces due to the formation of highly ordered interfacial structures. In these planar systems, mobility increases with pore size, eventually approaching *bulk*-like diffusivity in wider slits.

Overall, the results demonstrate that the curvature of the confining region is a critical factor in modulating both structural ordering and transport efficiency in confined ILs. These insights are especially relevant for the development of electrochemical devices, such as supercapacitors and batteries, which utilize nanoporous materials. Understanding how confinement geometry influences ionic layering, orientation, and diffusion provides a valuable framework for optimizing the architecture of porous electrodes

and enhancing the performance of energy storage systems. More broadly, this work reveals a fundamental principle: the curvature of the confining region not only modifies molecular organization but also can entirely reverse the dynamic behavior of the IL, even under structurally similar conditions.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for details on the polarizable forcefield; the structural parameters and molecular composition of the simulated systems; the simulation methodology fluxogram and the representation of characteristic structures, distances, and vectors of the simulated systems; and comparison of the constant potential and constant charge methods for 18 Å nanoslit.

ACKNOWLEDGMENTS

This work was part of Project No. CNS2023-144785, funded by MICIU/AEI/10.13039/501100011033 and the European Union “NextGenerationEU”/PRTR. Moreover, this work was funded by the Xunta de Galicia (Grant No. GRC ED431C 2024/06). This work was done within the framework of project HI_MOV – “Corredor Tecnológico Transfronterizo de Movilidad con Hidrógeno Renovable,” with reference 0160_HI_MOV_1_E, co-financed by the European Regional Development fund (ERDF), in the scope of Interreg VI A Spain – Portugal Cooperation Program (POCTEP) 2021-2027. M.O.L. thanks the Xunta de Galicia for his “Axudas de apoio á etapa predoutoral” (Grant No. ED481A 2022/236). H.M.C. and R.L.C. acknowledge their contracts under the framework of Project No. CNS2023-144785, funded by MICIU/AEI/10.13039/501100011033 and the European Union “NextGenerationEU”/PRTR. A.M.R. acknowledges the Ministry of Science, Innovation, and Universities for his FPU23/02994 grant. This publication and the contract of T.M.M. are part of Grant No. RYC2022-036679-I, funded by MICIU/AEI/10.13039/501100011033 and FSE+.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ángel Míguez-Roel: Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Martín Otero-Lema:** Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Raúl Lois-Cuns:** Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Hadrián Montes-Campos:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Trinidad Méndez-Morales:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal);

Project administration (lead); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Luis M. Varela:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in “Data associated to the publication: Study of the Properties of EMIMBF4 Under Nanoconfinement Conditions Using Polarizable Molecular Dynamics” at <http://doi.org/10.5281/zenodo.15790347>.

REFERENCES

- 1T. Welton, “Room-temperature ionic liquids. Solvents for synthesis and catalysis,” *Chem. Rev.* **99**, 2071–2084 (1999).
- 2S. A. Forsyth, J. M. Pringle, and D. R. MacFarlane, “Ionic liquids—an overview,” *Aust. J. Chem.* **57**, 113–119 (2004).
- 3N. V. Plechkova and K. R. Seddon, “Applications of ionic liquids in the chemical industry,” *Chem. Soc. Rev.* **37**, 123–150 (2008).
- 4T. L. Greaves and C. J. Drummond, “Protic ionic liquids: Properties and applications,” *Chem. Rev.* **108**, 206–237 (2008).
- 5D. R. MacFarlane, M. Forsyth, P. C. Howlett, M. Kar, S. Passerini, J. M. Pringle, H. Ohno, M. Watanabe, F. Yan, W. Zheng *et al.*, “Ionic liquids and their solid-state analogues as materials for energy generation and storage,” *Nat. Rev. Mater.* **1**, 1–15 (2016).
- 6M. J. Earle and K. R. Seddon, “Ionic liquids. Green solvents for the future,” *Pure Appl. Chem.* **72**, 1391–1398 (2000).
- 7G. A. O. Tiago, I. A. S. Matias, A. P. C. Ribeiro, and L. M. D. R. S. Martins, “Application of ionic liquids in electrochemistry—recent advances,” *Molecules* **25**, 5812 (2020).
- 8J. L. Anderson, J. Ding, T. Welton, and D. W. Armstrong, “Characterizing ionic liquids on the basis of multiple solvation interactions,” *J. Am. Chem. Soc.* **124**, 14247–14254 (2002).
- 9R. D. Rogers and K. R. Seddon, “Ionic liquids—solvents of the future?,” *Science* **302**, 792–793 (2003).
- 10V. I. Pârvulescu and C. Hardacre, “Catalysis in ionic liquids,” *Chem. Rev.* **107**, 2615–2665 (2007).
- 11D. Zhao, M. Wu, Y. Kou, and E. Min, “Ionic liquids: Applications in catalysis,” *Catal. Today* **74**, 157–189 (2002).
- 12A. Berthod, M. Ruiz-Angel, and S. Carda-Broch, “Ionic liquids in separation techniques,” *J. Chromatogr. A* **1184**, 6–18 (2008).
- 13B. Buszewski and S. Studzińska, “A review of ionic liquids in chromatographic and electromigration techniques,” *Chromatographia* **68**, 1–10 (2008).
- 14R. Kawano, H. Matsui, C. Matsuyama, A. Sato, M. A. B. H. Susan, N. Tanabe, and M. Watanabe, “High performance dye-sensitized solar cells using ionic liquids as their electrolytes,” *J. Photochem. Photobiol. A: Chem.* **164**, 87–92 (2004).
- 15P. Wang, S. M. Zakeeruddin, J.-E. Moser, and M. Grätzel, “A new ionic liquid electrolyte enhances the conversion efficiency of dye-sensitized solar cells,” *J. Phys. Chem. B* **107**, 13280–13285 (2003).
- 16A. Eftekhari, “Supercapacitors utilising ionic liquids,” *Energy Storage Mater.* **9**, 47–69 (2017).
- 17W. Zhou, M. Zhang, X. Kong, W. Huang, and Q. Zhang, “Recent advance in ionic-liquid-based electrolytes for rechargeable metal-ion batteries,” *Advanced Science* **8**, 2004490 (2021).
- 18M. Armand, F. Endres, D. R. MacFarlane, H. Ohno, and B. Scrosati, “Ionic-liquid materials for the electrochemical challenges of the future,” *Nat. Mater.* **8**, 621–629 (2009).
- 19H. Sakaebe, H. Matsumoto, and K. Tatsumi, “Application of room temperature ionic liquids to Li batteries,” *Electrochim. Acta* **53**, 1048–1054 (2007).
- 20M. Mezger, H. Schröder, H. Reichert, S. Schramm, J. S. Okasinski, S. Schöder, V. Honkimäki, M. Deutsch, B. M. Ocko, J. Ralston *et al.*, “Molecular layering of

fluorinated ionic liquids at a charged sapphire (0001) surface," *Science* **322**, 424–428 (2008).

- ²¹S. Perkin, "Ionic liquids in confined geometries," *Phys. Chem. Chem. Phys.* **14**, 5052–5062 (2012).
- ²²S. Zhang, J. Zhang, Y. Zhang, and Y. Deng, "Nanoconfined ionic liquids," *Chem. Rev.* **117**, 6755–6833 (2017).
- ²³L. Wang, T. Morishita, M. Toyoda, and M. Inagaki, "Asymmetric electric double layer capacitors using carbon electrodes with different pore size distributions," *Electrochim. Acta* **53**, 882–886 (2007).
- ²⁴P. Simon and Y. Gogotsi, "Materials for electrochemical capacitors," *Nat. Mater.* **7**, 845–854 (2008).
- ²⁵C. Largeot, C. Portet, J. Chmiola, P.-L. Taberna, Y. Gogotsi, and P. Simon, "Relation between the ion size and pore size for an electric double-layer capacitor," *J. Am. Chem. Soc.* **130**, 2730–2731 (2008).
- ²⁶F. Borghi, C. Piazzoni, M. Ghidelli, P. Milani, and A. Podestà, "Nanoconfinement of ionic liquid into porous carbon electrodes," *J. Phys. Chem. C* **125**, 1292–1303 (2021).
- ²⁷Z. Zhang, C. Li, A. Chandresh, and L. Heinke, "Conductivity measurement of ionic liquids confined in the nanopores of metal–organic frameworks: A case study for [bmim][tfsi] in hkust-1," *Ionics* **28**, 487–494 (2022).
- ²⁸C.-M. Wu and S.-Y. Lin, "Close packing existence of short-chain ionic liquid confined in the nanopore of silica ionogel," *J. Phys. Chem. C* **119**, 12335–12344 (2015).
- ²⁹A. Lewandowski and A. Świdarska-Moczek, "Ionic liquids as electrolytes for Li-ion batteries—An overview of electrochemical studies," *J. Power Sources* **194**, 601–609 (2009).
- ³⁰H. Adenusi, G. A. Chass, S. Passerini, K. V. Tian, and G. Chen, "Lithium batteries and the solid electrolyte interphase (SEI)—progress and outlook," *Adv. Energy Mater.* **13**, 2203307 (2023).
- ³¹J. M. Otero-Mato, H. Montes-Campos, O. Cabeza, L. J. Gallego, and L. M. Varela, "Nanoconfined ionic liquids: A computational study," *J. Mol. Liq.* **320**, 114446 (2020).
- ³²M. Foroutan, S. M. Fatemi, and F. Esmaeilian, "A review of the structure and dynamics of nanoconfined water and ionic liquids via molecular dynamics simulation," *Eur. Phys. J. E* **40**, 19 (2017).
- ³³S. Kondrat, G. Feng, F. Bresme, M. Urbakh, and A. A. Kornyshev, "Theory and simulations of ionic liquids in nanoconfinement," *Chem. Rev.* **123**, 6668–6715 (2023).
- ³⁴H.-Q. Liu, Y.-L. Wang, and B. Li, "Molecular insights into the nanoconfinement effect on the structure and dynamics of ionic liquids in carbon nanotubes," *Phys. Chem. Chem. Phys.* **26**, 14691–14704 (2024).
- ³⁵F. Federici Canova, H. Matsubara, M. Mizukami, K. Kurihara, and A. L. Shluger, "Shear dynamics of nanoconfined ionic liquids," *Phys. Chem. Chem. Phys.* **16**, 8247–8256 (2014).
- ³⁶M. Y. Ivanov, A. S. Poryvaev, D. M. Polyukhov, S. A. Prikhodko, N. Y. Adonin, and M. V. Fedin, "Nanoconfinement effects on structural anomalies in imidazolium ionic liquids," *Nanoscale* **12**, 23480–23487 (2020).
- ³⁷D. Bedrov, J.-P. Piquemal, O. Borodin, A. D. MacKerell, Jr., B. Roux, and C. Schröder, "Molecular dynamics simulations of ionic liquids and electrolytes using polarizable force fields," *Chem. Rev.* **119**, 7940–7995 (2019).
- ³⁸D. Bedrov, O. Borodin, Z. Li, and G. D. Smith, "Influence of polarization on structural, thermodynamic, and dynamic properties of ionic liquids obtained from molecular dynamics simulations," *J. Phys. Chem. B* **114**, 4984–4997 (2010).
- ³⁹M. E. Fisher and Y. Levin, "Criticality in ionic fluids: Debye-Hückel theory, Bjerrum, and beyond," *Phys. Rev. Lett.* **71**, 3826 (1993).
- ⁴⁰L. M. Varela, M. García, F. Sarmiento, D. Attwood, and V. Mosquera, "Pseudolattice theory of strong electrolyte solutions," *J. Chem. Phys.* **107**, 6415–6419 (1997).
- ⁴¹L. Varela, J. Carrete, M. García, L. Gallego, M. Turmine, E. Rilo, and O. Cabeza, "Pseudolattice theory of charge transport in ionic solutions: Corresponding states law for the electric conductivity," *Fluid Ph. Equilibria* **298**, 280–286 (2010).
- ⁴²H. Montes-Campos, S. Kondrat, E. Rilo, O. Cabeza, and L. M. Varela, "Random-alloy model for the conductivity of ionic liquid–solvent mixtures," *J. Phys. Chem. C* **124**, 11754–11759 (2020).
- ⁴³H.-S. Peng, "Study on the thermal decomposition of an ionic liquid propellant [EMIm][BF₄]," *Acta Astronaut.* **213**, 676–680 (2023).
- ⁴⁴J. Balasubramani, K. Gurusamy, and V. K. Balakrishnan, "Experimental investigation on the thermophysical properties of the ionic liquid-based binary system for heat transfer applications," *Therm. Sci.* **27**, 2977–2985 (2023).
- ⁴⁵P. Eastman, J. Swails, J. D. Chodera, R. T. McGibbon, Y. Zhao, K. A. Beauchamp, L.-P. Wang, A. C. Simmonett, M. P. Harrigan, C. D. Stern *et al.*, "OpenMM 7: Rapid development of high performance algorithms for molecular dynamics," *PLOS Comput. Biol.* **13**, e1005659 (2017).
- ⁴⁶K. Goloviznina, J. N. Canongia Lopes, M. Costa Gomes, and A. A. H. Pádua, "Transferable, polarizable force field for ionic liquids," *J. Chem. Theory Comput.* **15**, 5858–5871 (2019).
- ⁴⁷J. N. Canongia Lopes, J. Deschamps, and A. A. H. Pádua, "Modeling ionic liquids using a systematic all-atom force field," *J. Phys. Chem. B* **108**, 2038–2047 (2004).
- ⁴⁸J. N. Canongia Lopes, J. Deschamps, and A. A. H. Pádua, "Modeling ionic liquids using a systematic all-atom force field," *J. Phys. Chem. B* **108**, 11250 (2004).
- ⁴⁹J. N. Canongia Lopes and A. A. H. Pádua, "Molecular force field for ionic liquids composed of triflate or bistriflylimide anions," *J. Phys. Chem. B* **108**, 16893–16898 (2004).
- ⁵⁰J. N. Canongia Lopes and A. A. H. Pádua, "Molecular force field for ionic liquids III: Imidazolium, pyridinium, and phosphonium cations; chloride, bromide, and dicyanamide anions," *J. Phys. Chem. B* **110**, 19586–19592 (2006).
- ⁵¹J. N. Canongia Lopes, A. A. H. Pádua, and K. Shimizu, "Molecular force field for ionic liquids IV: Trialkylimidazolium and alkoxy carbonyl-imidazolium cations; alkylsulfonate and alkylsulfate anions," *J. Phys. Chem. B* **112**, 5039–5046 (2008).
- ⁵²K. Shimizu, D. Almantariotis, M. F. C. Gomes, A. A. H. Pádua, and J. N. Canongia Lopes, "Molecular force field for ionic liquids V: Hydroxyethylimidazolium, dimethoxy-2- methylimidazolium, and fluoroalkylimidazolium cations and bis(fluorosulfonyl)amide, perfluoroalkanesulfonylamide, and fluoroalkylfluorophosphate anions," *J. Phys. Chem. B* **114**, 3592–3600 (2010).
- ⁵³B. Jeziorski, R. Moszynski, and K. Szalewicz, "Perturbation theory approach to intermolecular potential energy surfaces of van der Waals complexes," *Chem. Rev.* **94**, 1887–1930 (1994).
- ⁵⁴E. G. Hohenstein and C. D. Sherrill, "Wavefunction methods for noncovalent interactions," *Wiley Interdiscip. Rev.: Comput. Mol. Sci.* **2**, 304–326 (2012).
- ⁵⁵L. Martínez, R. Andrade, E. G. Birgin, and J. M. Martínez, "PACKMOL: A package for building initial configurations for molecular dynamics simulations," *J. Comput. Chem.* **30**, 2157 (2009).
- ⁵⁶A. A. H. Padua, github.com/paduagroup/fftool, 2024.
- ⁵⁷A. A. H. Padua, github.com/paduagroup/pol_openmm, 2024.
- ⁵⁸H. Montes-Campos, T. Mendez-Morales, J. M. Otero-Mato, O. Cabeza, L. J. Gallego, E. Lomba, and L. M. Varela, "Ionic liquids nanoconfined in zeolite-templated carbon: A computational study," *J. Mol. Liq.* **318**, 114264 (2020).
- ⁵⁹J.-P. Ryckaert, G. Ciccotti, and H. J. C. Berendsen, "Numerical integration of the Cartesian equations of motion of a system with constraints: Molecular dynamics of *n*-alkanes," *J. Comput. Phys.* **23**, 327–341 (1977).
- ⁶⁰C. Y. Son, J. G. McDaniel, Q. Cui, and A. Yethiraj, "Proper thermal equilibration of simulations with Drude polarizable models: Temperature-grouped dual-Nosé-Hoover thermostat," *J. Phys. Chem. Lett.* **10**, 7523–7530 (2019).
- ⁶¹J. Åqvist, P. Wennerström, M. Nervall, S. Bjelic, and B. O. Brandsdal, "Molecular dynamics simulations of water and biomolecules with a Monte Carlo constant pressure algorithm," *Chem. Phys. Lett.* **384**, 288–294 (2004).
- ⁶²K.-H. Chow and D. M. Ferguson, "Isothermal-isobaric molecular dynamics simulations with Monte Carlo volume sampling," *Comput. Phys. Commun.* **91**, 283–289 (1995).
- ⁶³M. Otero-Lema, R. Lois-Cuns, M. A. Boado, H. Montes-Campos, T. Méndez-Morales, and L. M. Varela, "KUTE: Green-Kubo uncertainty-based transport coefficient estimator," *J. Chem. Inf. Model.* **65**, 3477–3487 (2025).
- ⁶⁴J. Yang, Z. Bo, H. Yang, H. Qi, J. Kong, J. Yan, and K. Cen, "Reliability of constant charge method for molecular dynamics simulations on EDLCs in nanometer and sub-nanometer spaces," *ChemElectroChem* **4**, 2486–2493 (2017).
- ⁶⁵L. Scalfi, M. Salanne, and B. Rotenberg, "Molecular simulation of electrode-solution interfaces," *Annu. Rev. Phys. Chem.* **72**, 189–212 (2021).
- ⁶⁶L. J. V. Ahrens-Iwers, M. Janssen, S. R. Tee, and R. H. Meißner, "Electrode: An electrochemistry package for atomistic simulations," *J. Chem. Phys.* **157**, 084801 (2022).

- ⁶⁷K. A. Dwelle and A. P. Willard, "Constant potential, electrochemically active boundary conditions for electrochemical simulation," *J. Phys. Chem. C* **123**, 24095–24103 (2019).
- ⁶⁸Z. Gong and A. A. H. Padua, "Effect of side chain modifications in imidazolium ionic liquids on the properties of the electrical double layer at a molybdenum disulfide electrode," *J. Chem. Phys.* **154**, 084504 (2021).
- ⁶⁹M. Shamsipur, A. A. M. Beigi, M. Teymouri, S. M. Pourmortazavi, and M. Irandoust, "Physical and electrochemical properties of ionic liquids 1-ethyl-3-methylimidazolium tetrafluoroborate, 1-butyl-3-methylimidazolium trifluoromethanesulfonate and 1-butyl-1-methylpyrrolidinium bis(trifluoromethyl sulfonyl)imide," *J. Mol. Liq.* **157**, 43–50 (2010).
- ⁷⁰A. Ghoufi, A. Szymczyk, and P. Malfreyt, "Ultrafast diffusion of ionic liquids confined in carbon nanotubes," *Sci. Rep.* **6**, 28518 (2016).
- ⁷¹Q. Berrod, F. Ferdeghini, P. Judeinstein, N. Genevaz, R. Ramos, A. Fournier, J. Dijon, J. Ollivier, S. Rols, D. Yu *et al.*, "Enhanced ionic liquid mobility induced by confinement in 1D CNT membranes," *Nanoscale* **8**, 7845–7848 (2016).
- ⁷²L. Gkoura, G. Diamantopoulos, M. Fardis, D. Homouz, S. Alhassan, M. Beazi-Katsioti, M. Karagianni, A. Anastasiou, G. Romanos, J. Hassan *et al.*, "The peculiar size and temperature dependence of water diffusion in carbon nanotubes studied with 2D NMR diffusion-relaxation D - $T_{2\text{eff}}$ spectroscopy," *Biomicrofluidics* **14**, 034114 (2020).
- ⁷³K. Hayamizu, S. Tsuzuki, S. Seki, and Y. Umebayashi, "Multinuclear NMR studies on translational and rotational motion for two ionic liquids composed of BF_4 anion," *J. Phys. Chem. B* **116**, 11284–11291 (2012).
- ⁷⁴L. Dick, K. Buchmüller, and B. Kirchner, "Coordination behavior of a confined ionic liquid in carbon nanotubes from molecular dynamics simulations," *J. Phys. Chem. B* **128**, 4472–4484 (2024).